

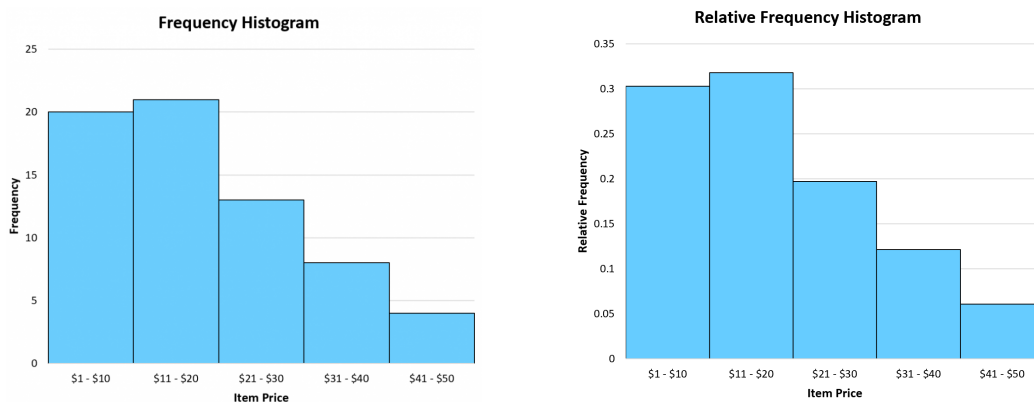
MAT 310, Handout 1: Introduction to Probability, Set Notation (1.1, 1.2, 2.1-2.3)

Goals of probability and statistics:

Relative frequency histogram

- intervals of equal width
- height of bars proportional to _____ of observations in the interval

Example:



Flip a coin. We know the probability of a Head, $P(H)$, is equal to $\frac{1}{2}$. But what does this mean?

Probability of an event (everyday language):

Probability Theory is a branch of mathematics that consists of rigorous methods for finding a number between 0 and 1 that agrees with the _____ of events occurring in a long sequence of random trials.

Example 1. Roll 2 dice. How could we calculate the probability that the sum is 8?

Sets

- In general, we will use capital letters for sets and lowercase letters for elements. For example, $A = \{a_1, a_2, a_3\}$.
- $\emptyset =$
- $\mathbb{S} =$

Set Operations

- $A \subseteq B$, i.e., A is a **subset** of B , implies every element in A _____.
Note that if $A \subseteq B$, then it is possible that $A = B$.
- $A \subset B$, i.e., A is a **proper subset** of B , implies every element in A is also in B , and also that _____.
- $A \cup B$, i.e., A **union** B , is the set of all elements _____.
- $A \cap B$, i.e., A **intersect** B , is the set of all elements _____.
- The **set difference** $A \setminus B$ (also denoted $A - B$) is the set of all elements _____.
- $\overline{A}$ (also denoted A^c), i.e., A **complement**, is the set of all elements _____.

- $A \cup \overline{A} =$
- $A \cap \overline{A} =$
- $A \subset B$ implies $A \cap B =$
- $A \subset B$ implies $A \cup B =$
- $A \subset B$ implies $A \cap \overline{B} =$
- If $A \cap B = \emptyset$, then we say that A and B are _____.

Distributive Laws

- $A \cap (B \cup C) =$

- $A \cup (B \cap C) =$

DeMorgan's Laws

- $\overline{A \cap B} =$

- $\overline{A \cup B} =$

Example 2. Out of 100 students, a total of 28 are taking Spanish, 26 are taking French, and 16 are taking German. Of these, 12 are taking both Spanish and French, 4 are taking both Spanish and German, and 6 are taking both French and German. Two students are taking all three languages.

- How many students are taking no languages?
- How many students are taking exactly one?
- How many students are in the set $F \cap \overline{G}$?
- How many students are in the set $\overline{S \cup F}$?